

Prob & Stats Assignment 02:

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Sec - 4H

Q. 8.11:-

$N=7$

$d=2$ $n=0,1,2$

$g=5$

$r=3$

$$a) P(0) = \frac{\binom{2}{0} \binom{5}{3}}{\binom{7}{3}} = \frac{2}{7}$$

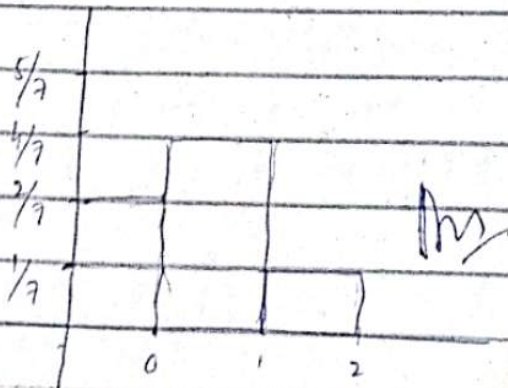
$$b) P(1) = \frac{\binom{2}{1} \binom{5}{2}}{\binom{7}{3}} = \frac{4}{7}$$

$$\binom{7}{3}$$

$$c) P(2) = \frac{\binom{2}{2} \binom{5}{1}}{\binom{7}{3}} = \frac{1}{7}$$

$$\binom{7}{3}$$

x	0	1	2
$P(x)$	$\frac{2}{7}$	$\frac{4}{7}$	$\frac{1}{7}$



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Qc) 3.18 - 3.20 :-

Soln

$$f(n) = \frac{2}{27} (1+n)$$

$$f(n) = \frac{2}{27} (1+n) \quad \text{or} \quad f(n) = \frac{2}{27} \left(\frac{n+n^2}{2} \right)$$

$$f(2) = \frac{2}{27} \left(\frac{2+2^2}{2} \right) = 8/27 \quad f(3) = \frac{2}{27} \left(\frac{3+3^2}{2} \right) = 15/27$$

$$f(4) = \frac{2}{27} \left(\frac{4+4^2}{2} \right) = 24/27$$

$$(a) P(2 < X < 4) = F(4) - F(2) = \frac{24}{27} - \frac{8}{27} = \frac{16}{27}$$

$$(b) P(3 \leq X < 4) = F(4) - F(3) = \frac{24}{27} - \frac{15}{27} = \frac{1}{3}$$

Qc) 3.13 :-

CDF values :-

$$f(n) = 0 \quad \text{for } n < 0$$

$$f(n) = 0.41 \quad 0 \leq n < 1$$

$$f(n) = 0.78 \quad 1 \leq n < 2$$

$$f(n) = 0.94 \quad 2 \leq n < 3$$

$$f(n) = 0.99 \quad 3 \leq n < 4$$

$$f(n) = 1 \quad n \geq 4$$

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Q#3.39.

Sol. $O=3, I=8, A=2, B=3, n=4$

$$P(x, y) = \binom{O}{x} \binom{A}{y} \binom{B}{n-x-y}$$

possible(x): 0, 1, 2, 3
possible(y): 0, 1, 2

$$\binom{L}{n}$$

(a)	x \ y	0	1	2
	0	0	$3/70$	$3/70$
	1	$3/70$	$18/70$	$9/70$
	2	$9/70$	$18/70$	$3/70$
	3	$3/70$	$18/70$	0

(b) $P[(x, y) \in A] \vee \{(x, y) \mid x+y \leq 2\}$

So possible (x, y) = (0, 0), (0, 1), (0, 2), (1, 0), (1, 1), (2, 0)

$$= 0 + 3/70 + 3/70 + 3/70 + 18/70 + 9/70$$

$$P(x+y \leq 2) = 36/70 \text{ Ans}$$

Q#3.49.

(a)

x	1	2	3
p(x)	0.1	0.35	0.55

(b)

y	1	3	5
p(y)	0.2	0.5	0.3

(c) $P(Y=3 \mid x=2) = \frac{0.10}{0.35} = 2/7$

[Signature]

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Q# 3-57r part(a)

Soln

2 find k.

$$\int_0^2 \int_0^1 \int_0^1 Kxy^2z \, dz \, dy \, dx = 1$$

$$= K \int_0^2 \int_0^1 \left[\frac{xy^2z^2}{2} \right]_0^1 dy \, dx = 1$$

$$= K \int_0^2 \left[\frac{y^3z}{6} \right]_0^1 dx = 1$$

$$= K \int_0^2 \frac{z}{6} dx = 1 \quad \text{or} \quad K \left[\frac{z^2}{12} \right]_0^2 = 1 \quad \frac{K}{3} = \boxed{K=3} \quad \text{Ans}$$

part(b) r $P\left(X < \frac{1}{4}, Y > \frac{1}{2}, 1/2 < Z < 2\right)$

Soln

$$P = \int_0^{1/4} \int_{1/2}^1 \int_1^2 3xy^2z \, dz \, dy \, dx$$

$$= 3 \int_0^{1/4} x \, dx \int_{1/2}^1 y^2 \, dy \int_1^2 z \, dz$$

$$= 3 \left(\frac{1}{32} \right) \left(\frac{7}{24} \right) \left(\frac{3}{2} \right) = 21/512 \quad \text{Ans}$$

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Q#3.62 r

(a) PMf r	n	1	3	5	7
	f(n)	0.4	0.2	0.2	0.2

(b) $P(1/2 < n < 7) = f(5) + f(7) = 0.4$ Ans

Q#4.1 r $E = \sum_{n=0}^4 n/f(n) \Rightarrow [E = 0.88]$ Ans

Q#4.7 r	Gain(n)	P(n)	$E = (4000)(0.3) + (-1000)(0.7)$
	4000	0.3	$[E = 500]$ Ans
	-1000	0.7	

Q#4.12 r

Sol. $E(n) = \int_0^1 n(2(1-n))dn$

$E(n) = 2 \int_0^1 (n - n^2) dn$

$E(n) = 2 \left[\frac{n^2}{2} - \frac{n^3}{3} \right]_0^1 dn$

$E(n) = 1/3$

$E[\text{profit}] = 1/3 \times 5000 \Rightarrow [E[\text{profit}] = \$1667]$ Ans

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Q# 4.20

$$[g(n)] = \int_0^{\infty} g(n) f(n) dn = \int_0^{\infty} e^{n/3} \cdot e^{-n} dn$$

Sol

$$\Sigma [e^{n/3}] = \int_0^{\infty} e^{-n/3} dn = \left[-3e^{-n/3} \right]_0^{\infty} = 3$$

Ans

Q# 4.34

$$E[X] = \Sigma x(p(x))$$

$$E[X] = (-2)(0.3) + (3)(0.2) + (5)(0.5)$$

$$E[X] = -0.6 + 0.6 + 2.5 = 2.5$$

$$E[X^2] = (-2)^2(0.3) + (3)^2(0.2) + (5)^2(0.5)$$

$$G = \sqrt{15.5 - 6.25} = 3.04$$

Ans

Q# 5.26

$$p = 0.6 \quad k = 6$$

$$n = 8 \quad \text{Sol}$$

$$P(6) = \binom{8}{6} (0.6)^6 (0.4)^2$$

$$P(6) = 0.209$$

Ans

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Q#458

Q#458

$$f(g(x)) = \int_{-\infty}^{\infty} g(x)f(x)dx$$

$$f(y) = \int_0^1 (60x^2 + 39x) \cdot x dx + \int_1^2 (60x^2 + 39x)(2-x) dx$$

$$= \int_0^1 (60x^2 + 39x) x dx \quad c) \left[\frac{60x^4}{4} + \frac{39x^3}{3} \right]_0^1 = 28$$

$$2) \int_1^2 (60x^2 + 39x)(2-x) dx = \int_1^2 (120x^2 + 78x) dx - \int_1^2 (60x^3 + 39x^2) dx$$

$$= \left[\frac{120x^3}{3} + \frac{78x^2}{2} \right]_1^2 - \left[\frac{60x^4}{4} + \frac{39x^3}{3} \right]_1^2$$

$$= 397 - (344 - 28) = 81 \quad \text{Ans}$$

$$= f(y) = 28 + 81$$

$$f(y) = 109$$

Ans

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Q45.121

$p = 0.25$ find $P(X \leq 4)$.
 $n = 3 \Rightarrow P(3) = \binom{9}{3} (0.25)^3 (0.75)^6 = 0.234$

$n = 2 \Rightarrow P(2) = \binom{9}{2} (0.25)^2 (0.75)^7 = 0.3$

$n = 1 \Rightarrow P(1) = \binom{9}{1} (0.25) (0.75)^8 = 0.225$

$n = 0 \Rightarrow P(0) = \binom{9}{0} (0.25)^0 (0.75)^9 = 0.075$

$P(X \leq 4) = 0.234 + 0.3 + 0.225 + 0.075 = 0.83$
 $P(X \leq 4) = 0.83$ Ans

Q45.571

$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}$

$\lambda = 2$

(a) $P(X \geq 4) = 1 - (X \leq 3)$

$P(0) = \frac{(e^{-2})(2)^0}{0!} = e^{-2}$

$P(1) = \frac{(e^{-2})(2)^1}{1!} = 2e^{-2} = 2e^{-2}$

$P(2) = \frac{(e^{-2})(2)^2}{2!} = 2e^{-2}$

$P(3) = \frac{(e^{-2})(2)^3}{3!} = \frac{4}{3}e^{-2}$

$P(X \geq 4) = 1 - \frac{19}{3}e^{-2} \Rightarrow 0.143$

(b) $P(X \leq 0) = e^{-2} = 0.135$ Ans

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Q1168.

$$\mu = 30, \sigma = 6$$

$$Z = \frac{x - \mu}{\sigma}$$

SDr

part(a). $Z = \frac{17 - 30}{6} = \frac{-13}{6} = -2.167$

c) $P(x > 17) = P(Z > -2.167)$

c) $P(Z > -2.167) = 1 - (P(Z < -2.167)) = 1 - 0.015 = 0.985$ Ans

part(b). $Z = \frac{22 - 30}{6} = \frac{-8}{6} = -1.333$

$(P(x < 22) = P(Z < -1.333) = 0.091)$ Ans

part(c). Area between $x = 32$ & $x = 41$

$$Z_1 = \frac{32 - 30}{6} = \frac{2}{6} = 0.333$$

$$Z_2 = \frac{41 - 30}{6} = \frac{11}{6} = 1.833$$

$$\begin{aligned} P(Z < 0.333) &= 0.63 \\ P(Z < 1.833) &= 0.967 \end{aligned} \quad \left\{ \begin{aligned} P(32 < x < 41) &= 0.967 - 0.63 \\ &= 0.337 \end{aligned} \right. \quad \text{Ans}$$

part(d). $Z_{0.5} \approx 0.841$

$$x = \mu + Z_{0.5} \sigma = 30 + (0.841)(6) = 35.05$$
 Ans

part(e). Middle 75% $\Rightarrow 1 - 0.75 = 0.25 \Rightarrow 12.5\%$

$$x_1 = \mu + Z_{0.125} \sigma = 30 + (-1.15)(6) = 30 - 6.9 = 23.1$$

$$x_2 = \mu + Z_{0.125} \sigma = 30 + (1.15)(6) = 30 + 6.9 = 36.9$$

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Q#6.11. $\mu = 200, \sigma = 15$

Sol $Z = \frac{x - \mu}{\sigma}$

(a) $Z = \frac{224 - 200}{15} = \frac{24}{15} = 1.6$

$P(x > 224) = P(Z > 1.6) = 1 - P(Z < 1.6)$
 $[P(Z > 1.6) = 1 - 0.945 = 0.055]$ Ans

part(h), $Z_1 = \frac{191 - 200}{15} = \frac{-9}{15} = -0.6$

$Z_2 = \frac{209 - 200}{15} = \frac{9}{15} = 0.6$

$P(191 < x < 209) = P(-0.6 < Z < 0.6) = 0.7257 - 0.2743$
 $= 0.4514$ Ans

part(i), $Z = \frac{230 - 200}{15} = \frac{30}{15} = 2$

$P(x > 230) = (P(Z > 2)) = 1 - 0.9772 = 0.0228$
 $(1000)(0.0228) = 22.8 \text{ days}$ Ans

part(ii), $Z = 0.25x - 0.674$

$x = \mu + Z\sigma = 200 + (-0.674)(15)$

$= 200 - 10.11 = 189.9$ Ans

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Q#1-2m

$$\mu = 10000 \quad \sigma = 100$$

part (a)

$$Z = \frac{10150 - 10000}{100} = \frac{150}{100} = 1.5$$

$$P(X > 10150) = 1 - P(X \leq 10150) = 1 - P(Z \leq 1.5) = 1 - 0.9332 = 0.0668$$

part (b) : 9800 - 10,200

(i) lower limit (9800)

$$Z = \frac{9800 - 10000}{100} = \frac{-200}{100} = -2$$

$$P(X < 9800) = P(Z < -2) \approx 0.0228$$

(ii) Upper limit (10,200)

$$P(X \geq 10,200)$$

$$\frac{10200 - 10000}{100} = \frac{200}{100} = 2$$

$$P(X > 10200) = P(Z > 2) = 1 - P(Z \leq 2) = 1 - 0.9772 = 0.0228$$

$$c) 0.0228 + 0.0228 = 0.0456$$

Total scrap